

ISE 555: Unit 3 Assignment

Fall 2026

For this assignment, all problems, including the writing assignment (Problem 5), are mandatory to receive full credit. A PDF file containing all your solutions as well as the writing assignment must be submitted on Brightspace by **4:00 pm on Monday, October 5th**. Make sure you have access to your submission during class on the day that the assignment is due. This will be helpful when presenting your solutions to the class in person.

1) Convert the following linear program to standard form:

$$\begin{array}{ll}\text{maximize} & z = 3x_1 + 5x_2 - 4x_3 \\ \text{subject to} & 7x_1 - 2x_2 - 3x_3 \geq 4 \\ & -2x_1 + 4x_2 + 8x_3 = -3 \\ & 5x_1 - 3x_2 - 2x_3 \leq 9 \\ & x_1 \geq 1, x_2 \leq 7, x_3 \geq 0\end{array}$$

2) Convert the following linear program to standard form:

$$\begin{array}{ll}\text{minimize} & z = x_1 - 5x_2 - 7x_3 \\ \text{subject to} & 5x_1 - 2x_2 + 6x_3 \geq 5 \\ & 3x_1 + 4x_2 - 9x_3 = 3 \\ & 7x_1 + 3x_2 + 5x_3 \leq 9 \\ & x_1 \geq -2, x_2 \text{ free}, x_3 \text{ free}\end{array}$$

3) Convert the following linear program to standard form:

$$\begin{array}{ll}\text{maximize} & z = 6x_1 - 3x_2 \\ \text{subject to} & 2x_1 + 5x_2 \geq 10 \\ & 3x_1 + 2x_2 \leq 40 \\ & x_1 \leq 15, x_2 \leq 15\end{array}$$

- 4) Write the following system of constraints in standard form and determine the basic solutions. Sketch the feasible region. Label each basic solution as either feasible or infeasible.

$$\begin{aligned}x_1 + x_2 &\leq 5 \\x_1 + 2x_2 &\leq 6 \\x_1 &\geq 0, x_2 \geq 0\end{aligned}$$

- 5) WRITING ASSIGNMENT: A company you work for has three locations and makes five different products. Each location can make all of the products, but they each do it at a different efficiency (i.e. one location may be more efficient making product XYZ but less efficient in making product ABC). In these tough economic times the corporate management has decided they need to close one of the locations. Write a brief paragraph explaining to the corporate management how you think your knowledge of optimization can be used to help them choose which location to close.

[NOTE: YOU DO NOT NEED TO WRITE ANY PROBLEM HERE – JUST DESCRIBE HOW OPTIMIZATION CAN BE USED]